
Litro Gas

Bay Management System

Live Demo Guide

 <https://litro-gas-production.up.railway.app>

What This System Does

The Litro Gas Bay Management System is a real-time Cylinder tracking , lorry tracking and bay assignment platform for LPG cylinder depots. When a lorry arrives at the facility, it is instantly assigned to the most efficient available bay reducing wait times and eliminating manual coordination.

- 8 bays managed simultaneously (4 Load + 4 Unload)
 - Real-time updates across all devices — no page refresh needed
 - Automatic bay assignment based on queue length
 - Full operation lifecycle: Arrival → Assignment → In Progress → Complete
 - Gate scanning via mobile phone — no dedicated hardware needed
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Quick Access Links

Page	URL
 Dashboard	https://litro-gas-production.up.railway.app
 QR Generator	https://litro-gas-production.up.railway.app/qr-generator
 Mobile Scanner	https://litro-gas-production.up.railway.app/mobile-scanner
 Connect Scanner	https://litro-gas-production.up.railway.app/mobile-connect

How to Run the Demo

Step 1 — Open the Dashboard

Open the Dashboard URL on a laptop or desktop browser. You will see 8 bay cards, live statistics, and a queue list. All bays start as Free (green).

Step 2 — Generate a Lorry QR Code

Navigate to the QR Generator page and fill in the lorry details:

- Registration Number (e.g. WP LA-1234)
- Selling Point / District (e.g. Colombo North)
- Operation Type → Load (filling) or Unload (returning empties)
- Target Cylinder Count (e.g. 120)

Click "Generate QR Code". A unique QR code is created for that lorry. Download it as an image.

Step 3 — Scan at the Gate (Mobile)

On your mobile phone, open the Mobile Scanner URL. Tap "Tap to Scan QR" and photograph the lorry's QR code. The system instantly reads it and assigns the lorry to the best available bay.

 **Tip:** If camera scanning does not work, tap "Manual Entry" to enter lorry details directly.

Step 4 — Watch the Dashboard Update

Switch back to the Dashboard. You will see the assigned bay change from Free → Occupied, the lorry details appear, and the Queue List update — all in real time.

Step 5 — Start the Operation

Click "Start" on an Occupied bay. The status changes to In Progress and a progress bar fills as cylinders are counted automatically.

Step 6 — Clear the Bay

Click "Clear" on a completed bay. If another lorry is waiting, it automatically moves in. Completed lorries appear in the Past Lorries log at the bottom of the dashboard.

Automatic Bay Assignment Algorithm

When a lorry QR code is scanned at the gate, the system performs the following logic automatically:

1. Reads the operation type from the QR code (Load or Unload)
2. Filters all bays that match that operation type
3. Checks which matching bay has the shortest current queue
4. Assigns the lorry to that bay immediately
5. If no bay is free, adds the lorry to the shortest queue and shows it on the dashboard

This eliminates the need for a person to manually direct lorries, reducing congestion and idle time at the facility.

System Architecture

Mobile Phone (Gate Scanner) → Node.js Server → Desktop Dashboard

- Frontend: React.js — runs in any modern browser
 - Backend: Node.js + Express — handles bay logic and state
 - Real-time: Socket.IO WebSockets — live push to all connected screens
 - QR Scanning: jsQR — pure JavaScript, fully compatible with iPhone Safari
 - Hosted on: Railway Cloud — 24/7 availability
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Tips for a Smooth Demonstration

- 💡 **Tip:** Before the demo, generate 2–3 lorry QR codes and save them to your phone gallery. Upload from gallery during the demo for a fast, smooth scan.
- 💡 **Tip:** Open the Dashboard on one screen and the Mobile Scanner on your phone at the same time to show the real-time sync effect live.
- 💡 **Tip:** Use the "Test Mode" button at the bottom of the Mobile Scanner page to simulate a lorry arrival without any QR code — perfect for a quick live demo.
- 💡 **Tip:** You can open the Dashboard in multiple browser tabs or on multiple laptops. All screens update simultaneously, showing real-time multi-user capability.